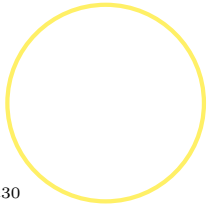
Sun $(0, 1.5e11)$



$2.0e30$

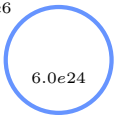
$G = 6.7e-11$

$F = \frac{GM_1M_2}{{D_{12}}^2}$

$\frac{GM_SM_E}{D_SD_S} = \frac{6.7e{-}11 \cdot 2.0e30 \cdot 6.0e24}{1.5e11 \cdot 1.5e11} = 3.6e22\ [kg\ m/s^2]$

$\frac{GM_S}{D_SD_S} = \frac{6.7e{-}11 \cdot 2.0e30}{1.5e11 \cdot 1.5e11} = 6.0e{-}3\ [m/s^2]$

$6.4e6$



$6.0e24$

Earth $(0, 0)$

$\frac{GM_MM_E}{D_MD_M} = \frac{6.7e{-}11 \cdot 7.4e22 \cdot 6.0e24}{3.8e8 \cdot 3.8e8} = 2.1e20\ [kg\ m/s^2]$

$\frac{GM_M}{D_MD_M} = \frac{6.7e{-}11 \cdot 7.4e22}{3.8e8 \cdot 3.8e8} = 3.4e{-}5\ [m/s^2]$

$7.4e22$

Moon $(3.8e8, 0)$